

Unit 1, Lesson 6: Transformations on the Plane – Part 2



Benchmark

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.



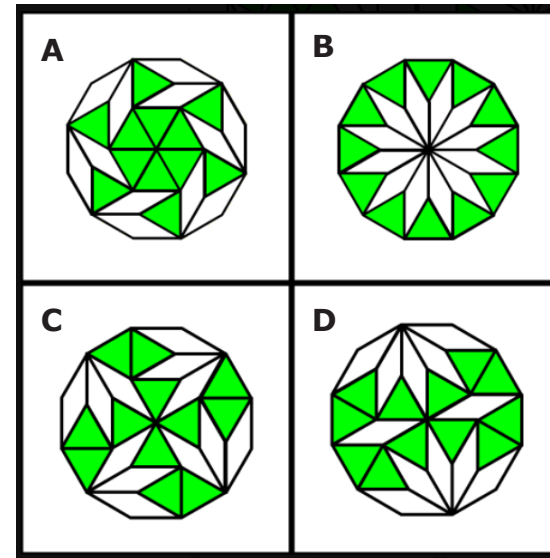
Learning Target

- ☐ I can describe a single transformation that was applied.



1.6.1 Warm-Up: Which One Doesn't Belong?

1. The diagram shows four polygons.

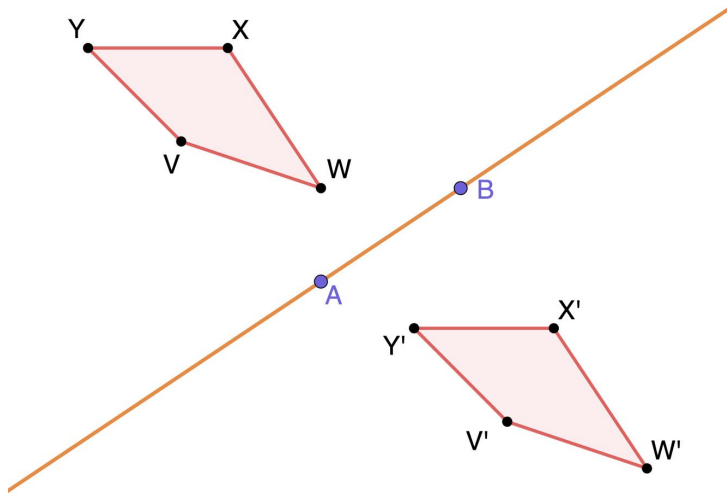


- a. Which polygon does not belong?
- b. Describe your reasoning.



1.6.2 Guided Instruction: Determining a Single Transformation

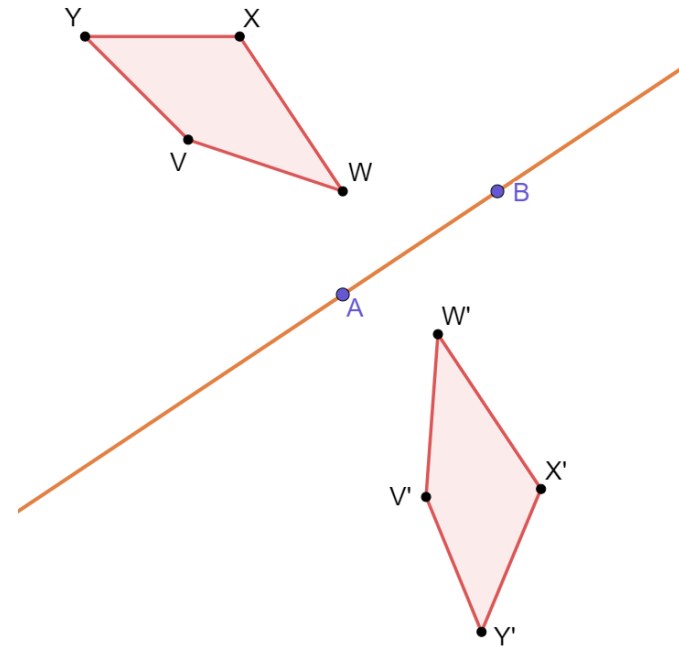
1. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.



Describe the single transformation that was used to create $V'W'X'Y'$.

Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	

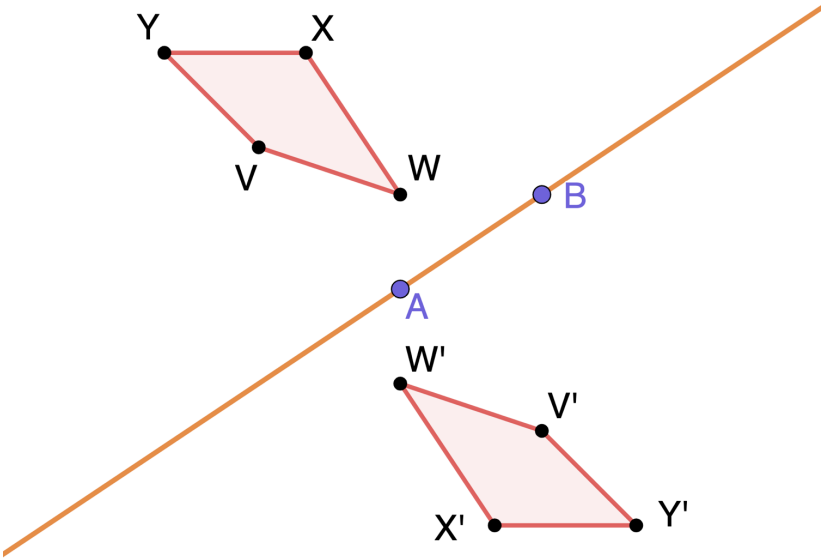
2. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.



Describe the single transformation that was used to create $V'W'X'Y'$.

Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	

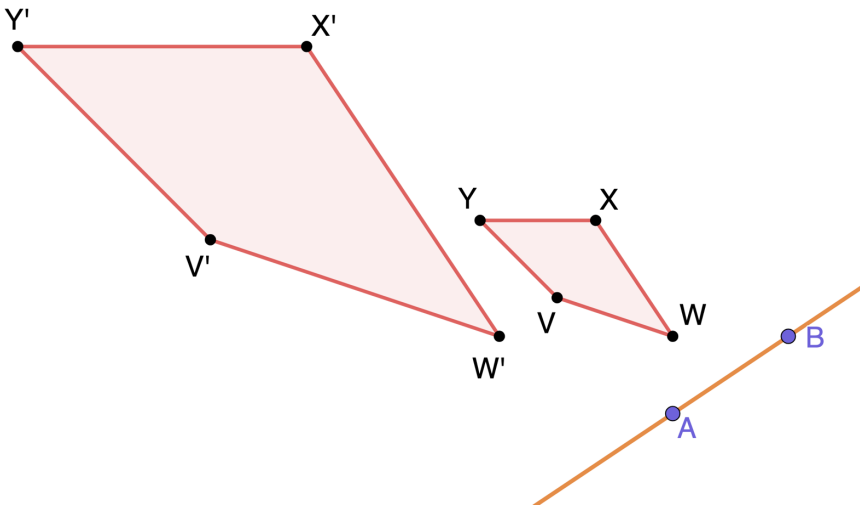
3. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.



Describe the single transformation that was used to create $V'W'X'Y'$.

Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	

4. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.



Describe the single transformation that was used to create $V'W'X'Y'$.

Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	



5. Complete the chart so it can be used to distinguish between similar-looking transformations. First, write your own hints. Then, exchange hints with your partner.

Transformation	My Hints	My Partner's Hints
Dilation		
Reflection		
Rotation		
Translation		

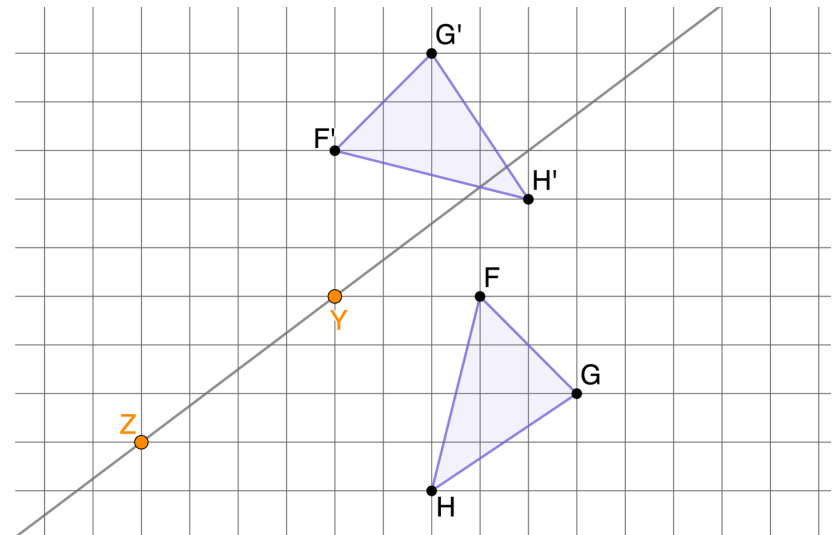
6. Which transformations resulted in congruent figures?
- ☐ Dilation
 - ☐ Reflection
 - ☐ Rotation
 - ☐ Translation
7. When a transformation results in a congruent figure, what is another name that can be used for that transformation?



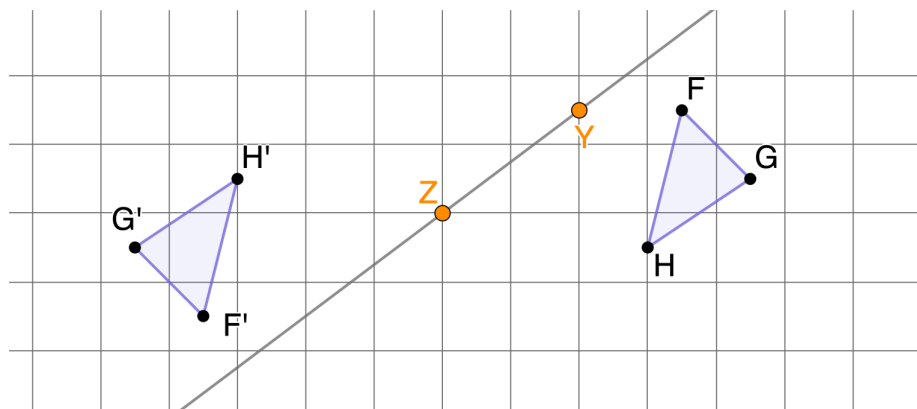
1.6.3 Guided Practice: Describing Transformations

Each of the diagrams shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.

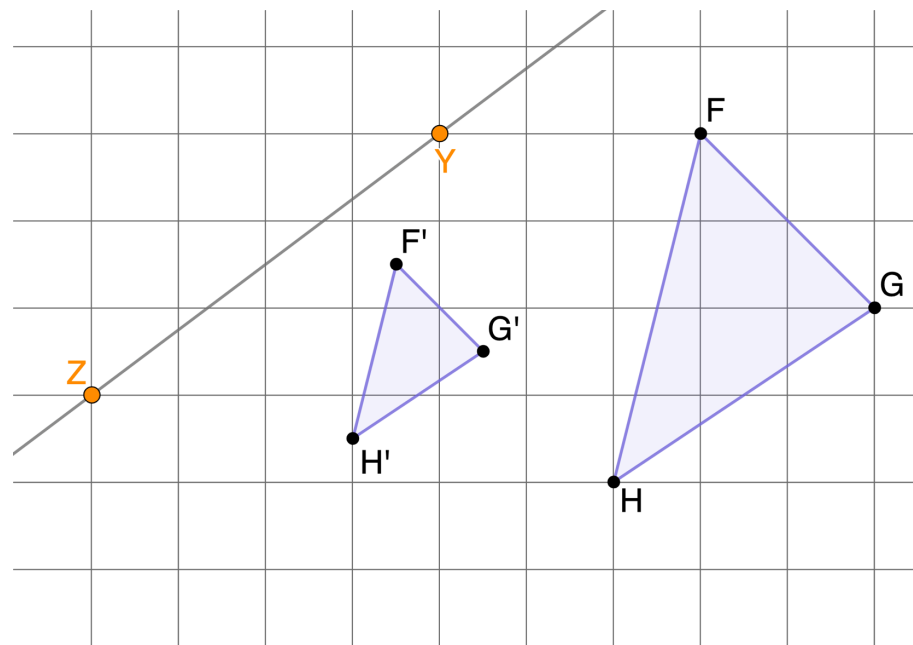
1. Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.



2. Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.



3. Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.

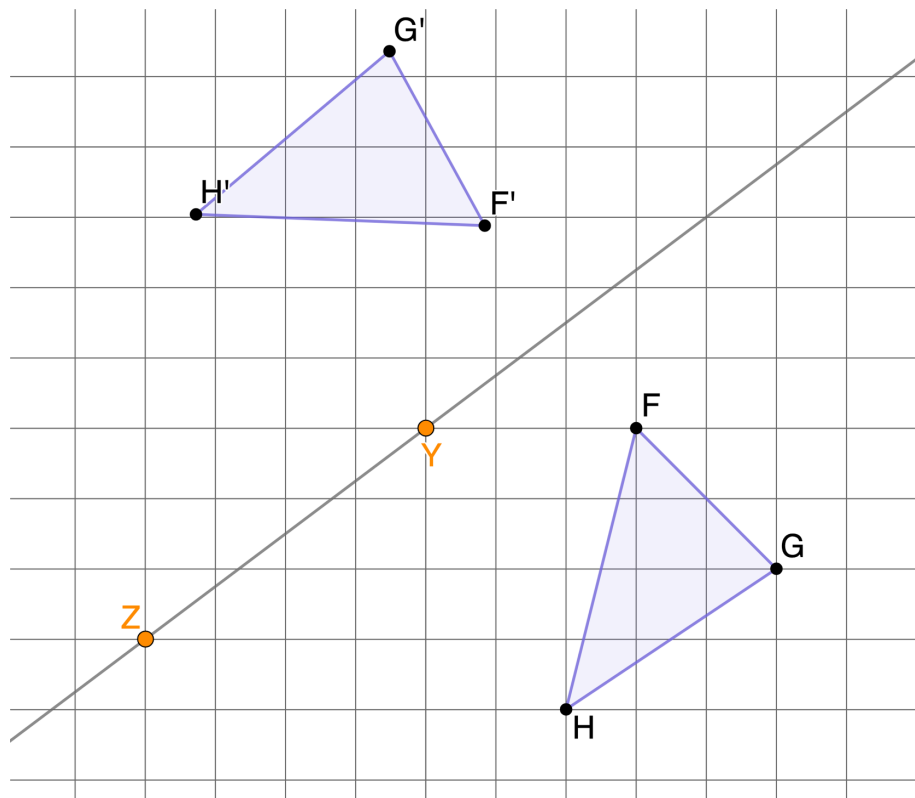


4. Review your hints chart and add anything else you think will be helpful.



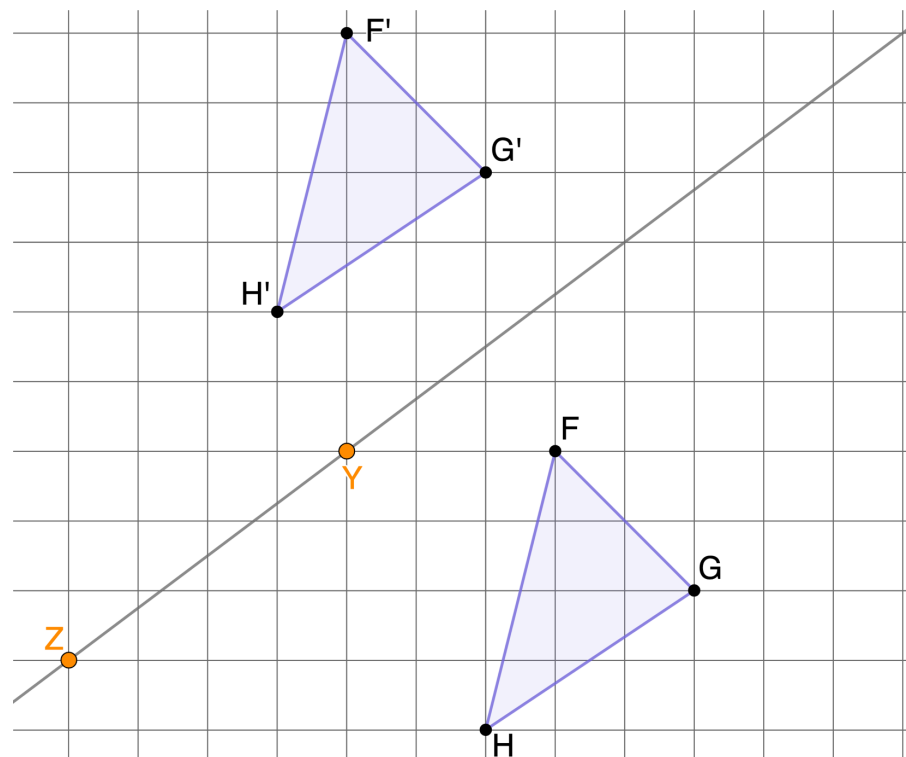
1.6.4 Your Turn!

1. The diagram shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.



Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.

2. The diagram shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.

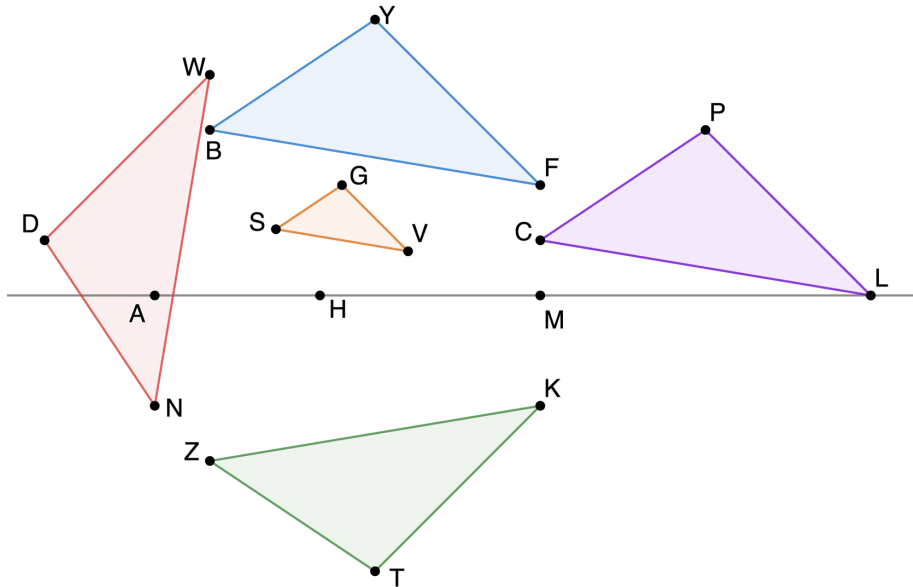


Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.



1.6.5 Wrap-Up: Ticket Out the Door

- The diagram shows four different single transformations of $\triangle YFB$.



Complete the chart by naming and describing the type of transformation that was applied to $\triangle YFB$ to result in each triangle.

Triangle	Type of Transformation	Description of the Transformation
$\triangle PLC$		
$\triangle TKZ$		
$\triangle GVS$		
$\triangle DWN$		

Unit 1, Lesson 7: Congruence and Similarity



Benchmarks

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.

MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar.



Learning Targets

- ☐ I can recognize the relationship between corresponding sides and corresponding angles of congruent and similar figures.
- ☐ I can write congruence statements to relate corresponding parts following a rigid transformation.
- ☐ I can write similarity statements to relate corresponding parts following a dilation.